Topic 7 Differential Equations

- 7.1 Introduction to Differential Equations 524
- 7.2 Separable Differential Equations 544
- 7.3 Applications of Separable Equations 566

7.1 Introduction to Differential Equations

Example 7.1

Baby galahs gain mass at a rate proportional to the difference between the average adult weight for the species and the current weight of the baby bird. This rate of mass increase is given by

$$M'(t) = \frac{1}{5}(300 - M(t))$$

where $M(t)$ denote the mass of a bird at time t (in months after it hatches).

Questions:

- Do we have enough information to find $M(t)$?
- How do we find $M(t)$?
- Is $M(t)$ unique?

7.1 Introduction to Differential Equations

An equation involving a variable (say x), an unknown function (say y), and the derivatives of y with respect to x:

$$x,\ y,\ \frac{dy}{dx},\ \frac{d^2y}{dx^2},\ \ldots,\ \frac{d^ny}{dx^n}$$

is called a <u>differential equation</u>. The order of the highest derivative in the differential equation, is called the <u>order</u> of the differential equation.

Example 7.2

$\frac{dy}{dx} = -5y$	order =
$\frac{d^3y}{dt^3} + t\frac{dy}{dt} + (t^4 - 1)y = \sin(t)$	order =
$e^x f'(x) + f''(x) = f(x)^3 + x$	order =

7.1 Introduction to Differential Equations: Solutions

A differential equation is an equation made of functions. So it is reasonable to expect that its solution will be a function.

Example 7.3

Consider the following equation where y is a function of x:

$$y + \cos^2(x) = x^2 + 1.$$

1. Is $y = \sin^2(x) + x^2$ a solution?
2. Is $y = x^2$ a solution?

7.1 Introduction to Differential Equations: Solutions

Definition 7.4

A <u>solution</u> to a differential equation is a function such that when substituted into both sides of the differential equation, equality results.

7.1 Introduction to Differential Equations: Solutions

Example 7.5

Verify that $y = e^{3x}$ is a solution of the differential equation:

$$\frac{d^2y}{dx^2} = 15y - 2\frac{dy}{dx}.$$

Homework 68

Use implicit differentiation to verify that any function y for which

$$\log(y) = xy^2 + C$$

is a solution of the differential equation

$$\frac{dy}{dx} = \frac{y^3}{1 - 2xy^2}.$$

7.1 Introduction to Differential Equations: Solutions

Homework 69

Find the constants a, b and c such that $y = a + bx + cx^2$ *is a solution of the differential equation*

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 4y = 4x^2.$$

[Hint: Substitute the formula for y into both sides of the equation and equate coefficients.]

7.1 Introduction to Differential Equations: Solutions

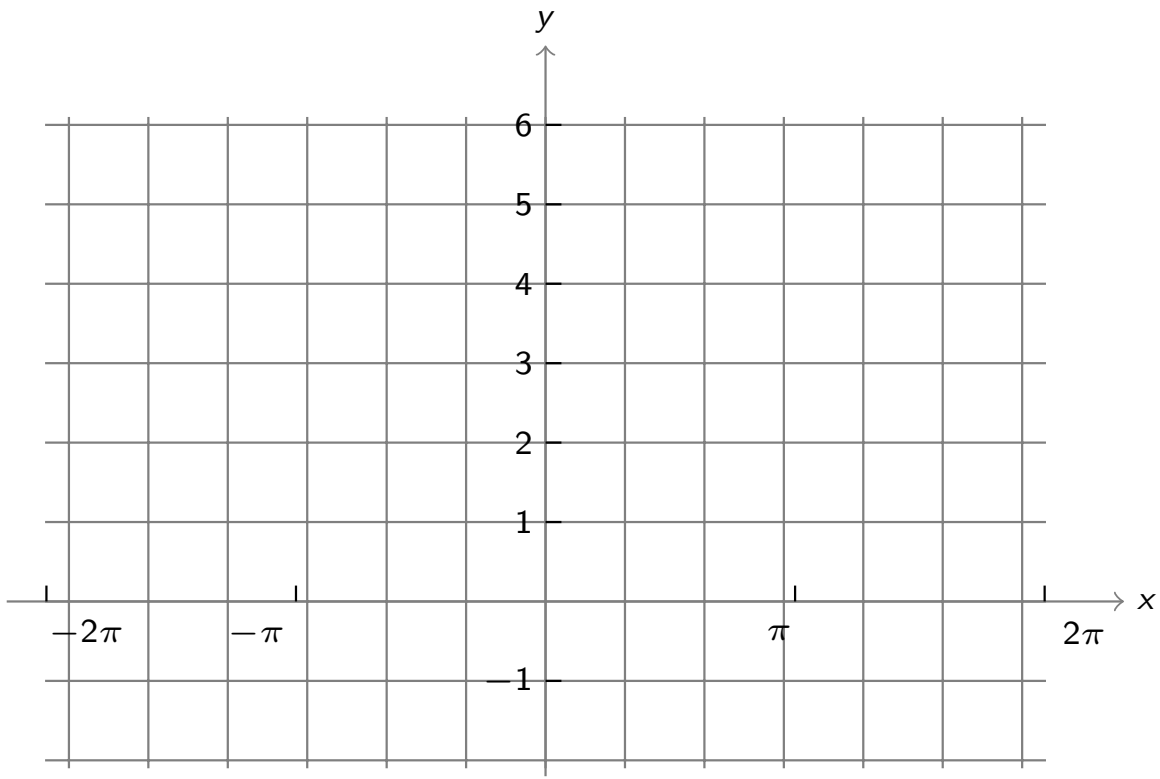
Example 7.6

By integrating on both sides, find a formula for all solutions of the following differential equation :

$$\frac{dy}{dx} = \cos(x).$$

7.1 Introduction to Differential Equations: Solutions

- Sketch the graphs of some typical solutions of $\dfrac{dy}{dx} = \cos(x)$.



7.1 Introduction to Differential Equations: Solutions

- A typical differential equation has infinitely many solutions.
- The solutions typically have similar formulas, differing only by the presence of some constants.

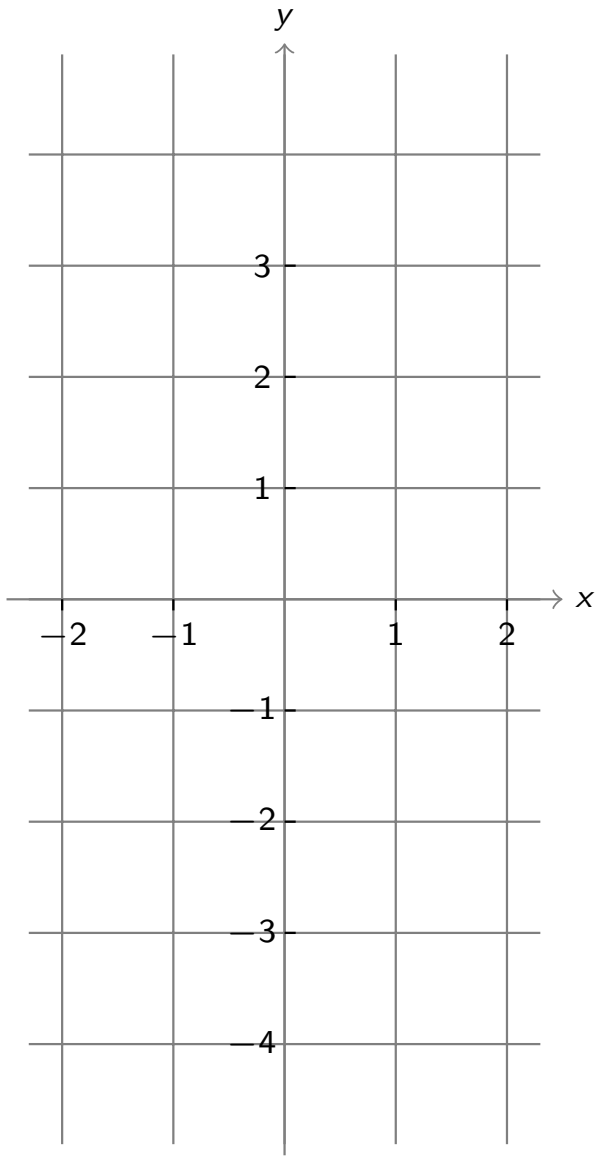
7.1 Introduction to Differential Equations: Solutions, General and Particular

Example 7.7

Verify $f(x) = x^2 + \frac{C}{x}$ is a solution of the differential equation $f'(x) + \frac{1}{x}f(x) = 3x$ for all values of C. Sketch solutions with $C = -2, -1, 0, 1, 2$.

7.1 Introduction to Differential Equations: Solutions, General and Particular

7.1 Introduction to Differential Equations: Solutions, General and Particular



7.1 Introduction to Differential Equations: Solutions, General and Particular

Example 7.8

Let $C > 0$. Verify $M(t) = -Ce^{-\frac{1}{5}t} + 300$ is a solution of the differential equation

$$M'(t) = \frac{1}{5}(300 - M(t))$$

7.1 Introduction to Differential Equations: Solutions, General and Particular

Definition 7.9

The general solution of a differential equation is a formula for all of the possible solutions of the differential equation. The general solution describes the set of all possible solutions to the differential equation.

The general solution to the differential equation $\frac{dy}{dx} = \cos(x)$ is $y = \sin(x) + C$.

Fixing any value for C gives us a particular solution.

7.1 Introduction to Differential Equations: Solutions, General and Particular

Example 7.10

Give a formula for $M(t)$ from Example 7.1 when $M(0) = 15g$.

7.1 Introduction to Differential Equations: Solutions, General and Particular

Definition 7.11

A particular solution to a differential equation arises from a general solution by choosing a value for the constant(s) of integration.

We decide how to fix the constant(s) of integration based on other information, usually called initial condition(s).

A differential equation together with initial conditions is called an initial value problem

7.1 Introduction to Differential Equations: Solutions, General and Particular

Problem: All we have really done so far is talking about verifying general solutions, finding particular solutions and solving initial value problems for very straightforward differential equations.

Is there a general method for solving all initial value problems?

7.1 Introduction to Differential Equations: Solutions, General and Particular

No.
(See MAST20009 – Differential Equations)

However, there are some others that are straightforward to solve. In the subsequent section we discuss one technique for solving differential equations.

7.2 Separable Differential Equations

Example 7.12

Find the general solution for the differential equation $\dfrac{dy}{dx} = \frac{x^2+1}{y^2}$

7.2 Separable Differential Equations

Definition 7.13

Differential equations of the form

$$\frac{dy}{dx} = F(x)G(y)$$

are called <u>separable</u>.

7.2 Separable Differential Equations: Method

- To solve a separable differential equation

$$\frac{dy}{dx} = F(x)G(y).$$

  we first divide both sides by $G(y)$ (assuming $G(y) \neq 0$).
- This step, often called separating the variables, gives

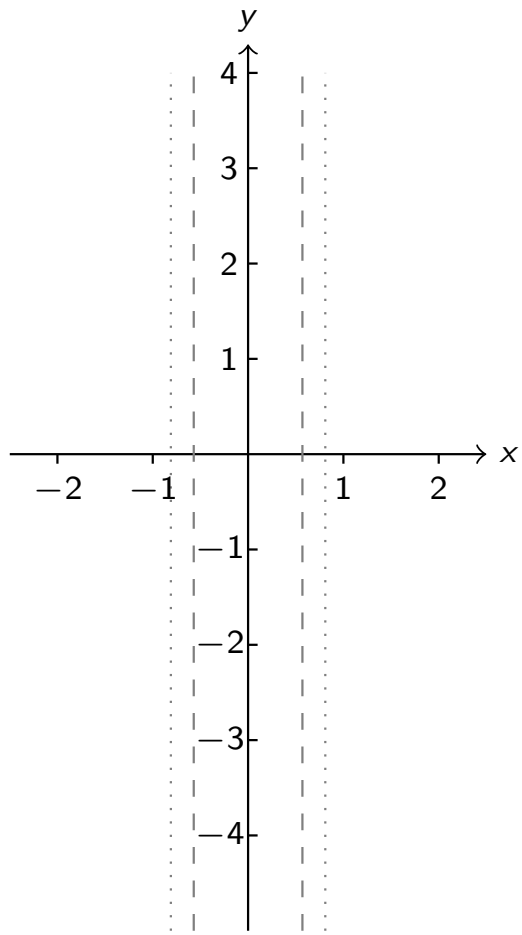
$$\frac{1}{G(y)}\frac{dy}{dx} = F(x)$$

- We then attempt to antidifferentiate both sides with respect to x.
- If possible, we solve the resulting function equation to obtain a formula for y (containing a constant of integration).
- (We will consider the situation where $G(y) = 0$ later.)

Example 7.14

Assuming $y \neq 0$, find the general solution of

$$\frac{dy}{dx} = -6xy^2.$$



7.2 Separable Differential Equations

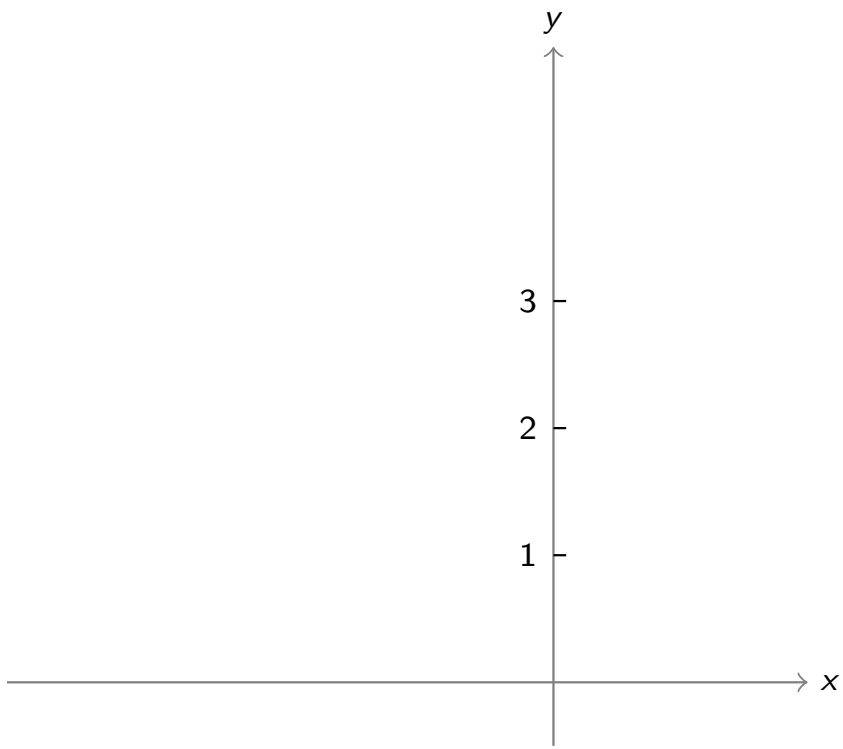
Example 7.15

Find the general solution of the following differential equation and sketch some typical solutions, assuming $y > 0$.

$$\frac{dy}{dx} = y.$$

This is a separable differential equation in disguise!

7.2 Separable Differential Equations



7.2 Separable Differential Equations: Method

- Our method for solving separable differential equations can fail at various steps:
  1. We may not be able to find a formula for $\int \frac{1}{G(y)}\,\mathrm{d}y$
  2. We may not be able to find a formula for $\int F(x)\,\mathrm{d}x$
  3. After integrating both sides, we may not be able to solve to obtain a formula for y as a function of x.

Example 7.16

Can we solve the following separable differential equation?

$$\frac{dy}{dx} = \cos(x^2)e^{-y^2}$$

7.2 Separable Differential Equations: Method

Example 7.17

Can we solve the following separable differential equation?

$$\frac{dy}{dx} = \frac{1}{6y^5 - 2y + 1}$$

7.2 Separable Differential Equations

- Some differential equations require a little rearrangement to put them into separable form.

Example 7.18

Show that the following differential equation is separable:

$$\frac{dy}{dx} = y^2 \sin(x) + y^2 - 4\sin(x) - 4$$

7.2 Separable Differential Equations: Method

Homework 70

Solve the differential equation of Example 7.18. You will need to apply partial fractions to integrate the LHS. You may assume that $y > 2$.

7.2 Separable Differential Equations

Homework 71

Solve the initial value problem:

$$\frac{1}{3x^2y}\frac{dy}{dx} = \frac{1}{\sqrt{x^3-11}} \qquad \textit{where} \quad y = 1 \textit{ when } x = 3.$$

7.2 Separable Differential Equations: Method

- To solve a separable differential equation

$$\frac{dy}{dx} = F(x)G(y).$$

  we first divide both sides by $G(y)$, assuming $G(y) \neq 0$.
- This step, often called separating the variables, gives

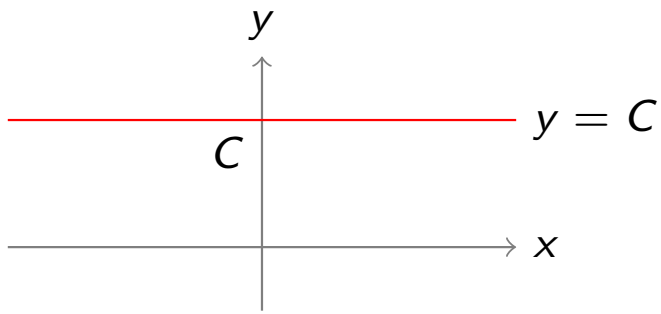
$$\frac{1}{G(y)}\frac{dy}{dx} = F(x)$$

- We then attempt to antidifferentiate both sides with respect to x.
- If possible, we solve the resulting function equation to obtain a formula for y (containing a constant of integration).
- (We will consider the situation where $G(y) = 0$ later)

7.2 Separable Differential Equations: Constant Solutions

- For any y value such that $G(y) = 0$, a separable differential equation
$$\frac{dy}{dx} = F(x)G(y)$$
collapses to
$$\frac{dy}{dx} = 0.$$
which can be antidifferentiated:
$$\int \frac{dy}{dx}\,\mathrm{d}x = \int 0\,\mathrm{d}x \qquad \Rightarrow \qquad y = C.$$

- The solution is always a constant function.



7.2 Separable Differential Equations: Constant Solutions

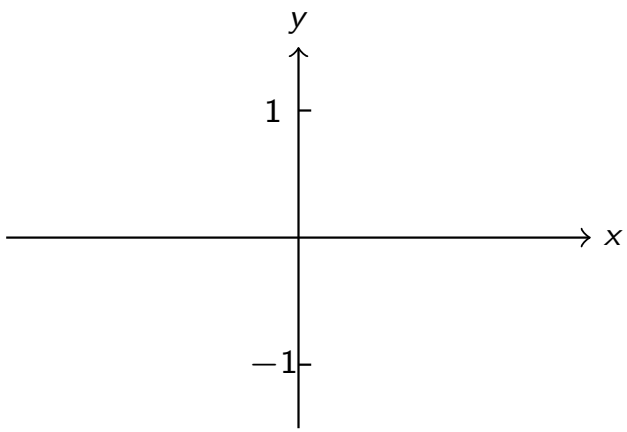
Example 7.19

Solve the initial value problem

$$\frac{dy}{dx} = -6xy^2 \qquad \text{where} \qquad y = 0 \text{ when } x = 0.$$

and sketch the solution.

7.2 Separable Differential Equations: Constant Solutions



7.2 Separable Differential Equations: Constant Solutions

- For separable differential equations constant solutions are typically straightforward to find. We solve the equation $G(y) = 0$.

Example 7.20

Find any constant solutions of the following separable differential equations

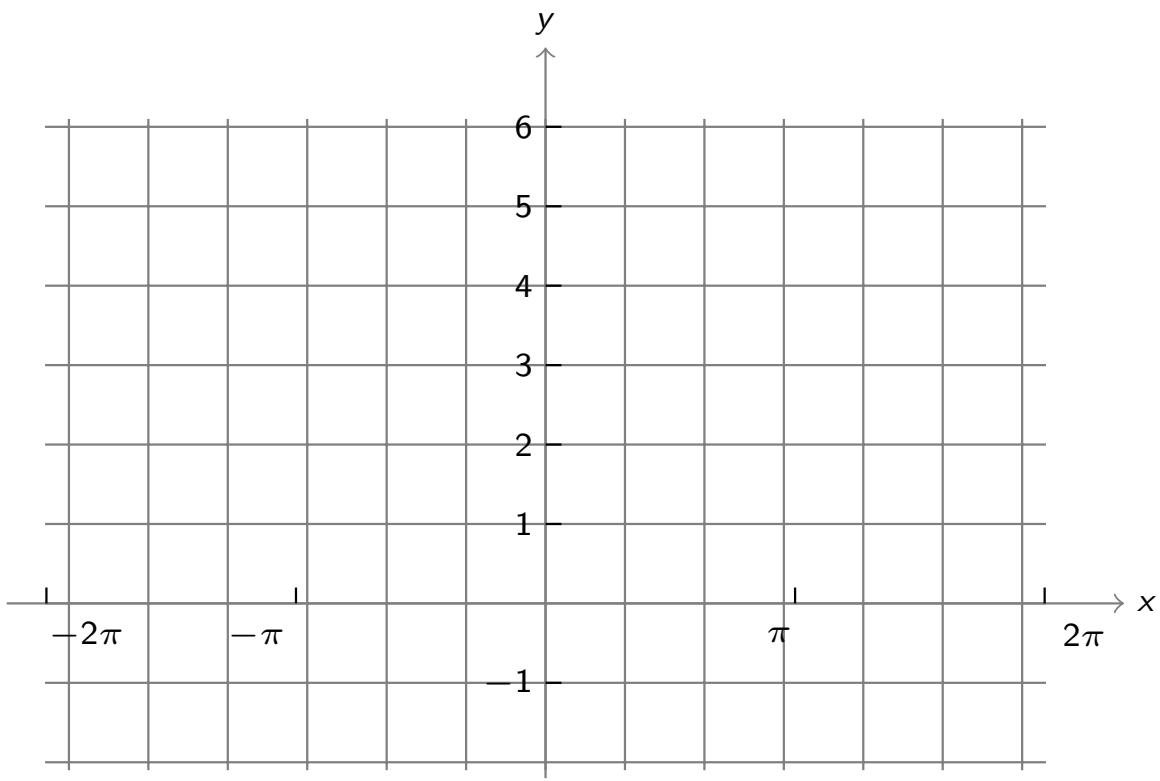
1. $\dfrac{dy}{dx} = y - \dfrac{y^2}{4}$.
2. $\dfrac{dy}{dx} = e^y \sin(x)$.
3. $\dfrac{dy}{dx} = \cos^2(y)e^x$.
4. $\dfrac{dy}{dx} = y$.
5. $\dfrac{dy}{dx} = y^{\frac{1}{3}}$.

7.2 Separable Differential Equations: Constant Solutions

7.2 Separable Differential Equations: Unique Solutions

In our work in solving initial value problems, our final answer has always been a single function.

- Is it possible for there to be more the one solution to a initial value problem with a separable differential equation?
- Is it possible for solution curves to cross?



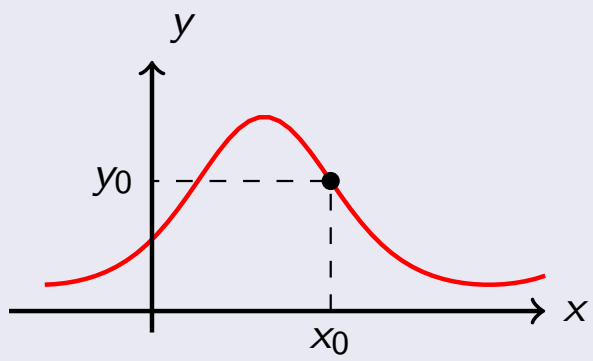
7.2 Separable Differential Equations: Unique Solutions

Theorem 7.21 (Uniqueness)

Suppose F is continuous and defined at x_0 and G' is continuous and defined at y_0. Then the initial value problem:

$$\frac{dy}{dx} = F(x)G(y) \qquad y = y_0 \text{ when } x = x_0$$

has a unique solution on some interval I containing x_0.



- Distinct solution curves can't touch (or cross)!

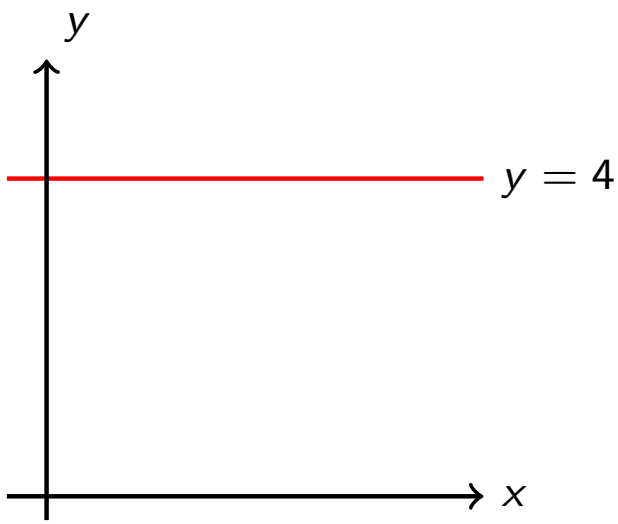
7.2 Separable Differential Equations: Unique Solutions

Example 7.22

Use the constant solutions found in Example 7.20(1) and Theorem 7.21 to find bounds for the range of the solution of the initial value problem:

$$\frac{dy}{dx} = y - \frac{y^2}{4} \qquad \text{where} \quad y = 1 \text{ when } x = 0.$$

7.2 Separable Differential Equations: Unique Solutions



7.2 Separable Differential Equations

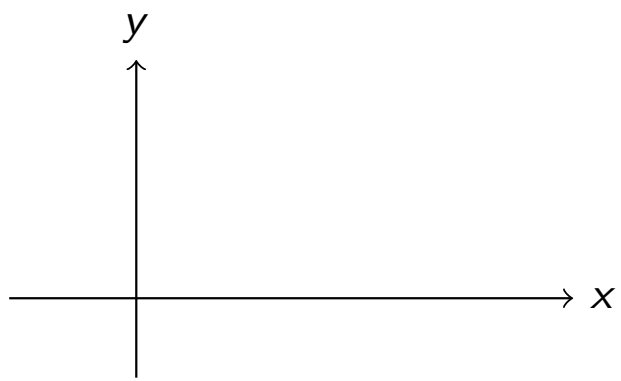
Example 7.23

Consider the differential equation

$$\frac{dy}{dx} = y^{\frac{1}{3}}.$$

1. Find the constant solutions.
2. Find the general solution.
3. How many solution curves pass through the point $(0, 0)$?
4. Why does this not contradict Theorem 7.21?

7.2 Separable Differential Equations



7.3 Applications of Differential Equations

We finish with some applications designed to illustrate the spread of applications of separable differential equations.

- Exponential growth
- Logisitic growth
- Heating and Cooling
- Tree Growth

7.3 Applications of Differential Equations: Exponential Growth

Example 7.24

The rate of growth of a certain mouse population is proportional to the current number of mice present:

$$\frac{dM}{dt} = 0.2M,$$

where M is the number of mice t weeks after observation begins. There are initially 50 mice.

1. Explain why $M(t) > 0$ for all t.
2. Find a formula for the number of mice $M(t)$ at time t.
3. When will the mouse population reach 500?

7.3 Applications of Differential Equations: Exponential Growth

7.3 Applications of Differential Equations: Exponential Growth

Homework 72

A certain culture of bacteria is growing at a rate proportional to the current number B of bacteria present. If initially there are 1000 bacteria and after 1 hour this number has doubled:

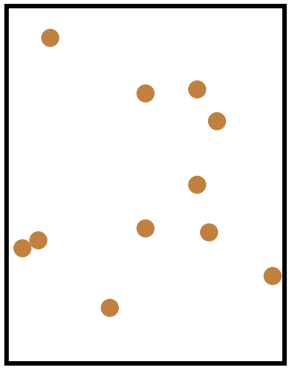
1. *Find the number of bacteria $B(t)$ present at any time t.*
2. *How many bacteria are present after $1\frac{1}{2}$ hours?*
3. *When will the number of bacteria reach 1 million?*

7.3 Applications of Differential Equations: Logistic Growth

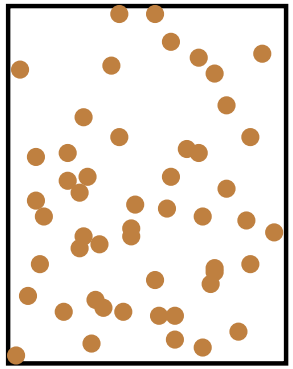
- In the long term, no population can grow exponentially, so the exponential growth model
$$\frac{dP}{dt} = kP$$
eventually breaks down.
- Adding an extra term gives a logistic growth differential equation:
$$\frac{dP}{dt} = kP - rP^2.$$
- The term $-rP^2$ gives a reduced growth rate due to overcrowding
$$\frac{dP}{dt} = kP\left(1 - \frac{r}{k}P\right).$$
- But why should the effects of overcrowding be proportional to the square of the population size?

7.3 Applications of Differential Equations: Law of Mass Action

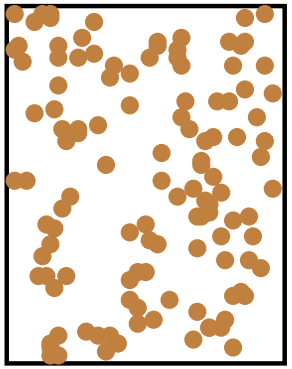
- The simplest model expresses overcrowding in terms of the probability of encounters between individuals.



Small population, few encounters.



Medium population, more encounters.



Large population, many encounters.

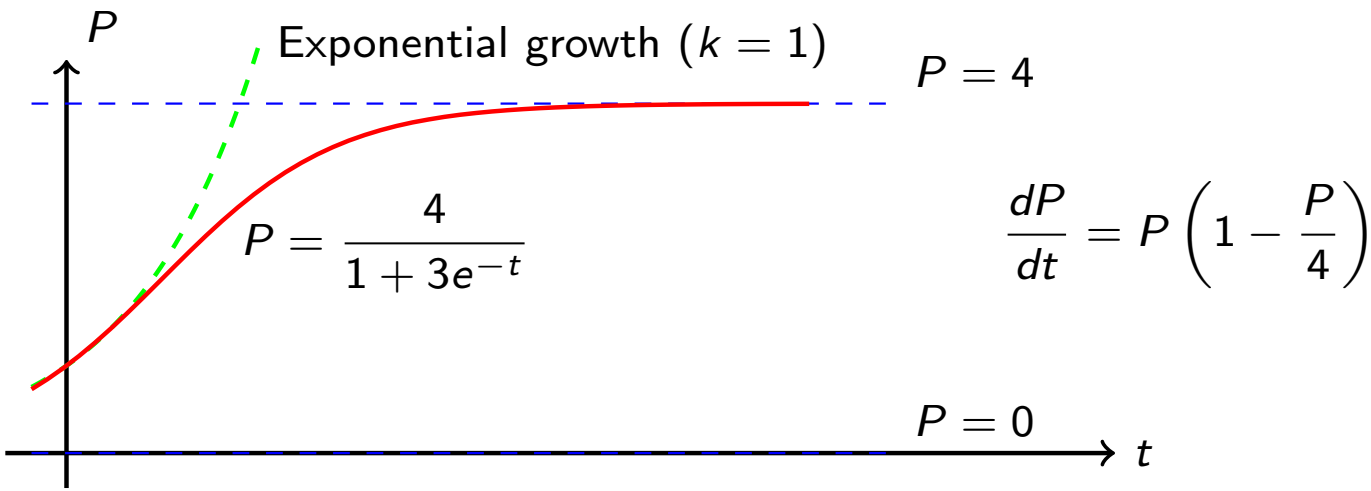
- Assuming individuals are placed randomly, this should be proportional to the number of pairs of distinct individuals.
- In a population of size P, the number of pairs of distinct individuals is approximately $\frac{1}{2}P^2$.
- Hence we model effects of overcrowding as proportional to P^2.
- This idea is similar to the law of mass action in chemistry.

7.3 Applications of Differential Equations: Logistic Growth

- The general logistic growth differential equation

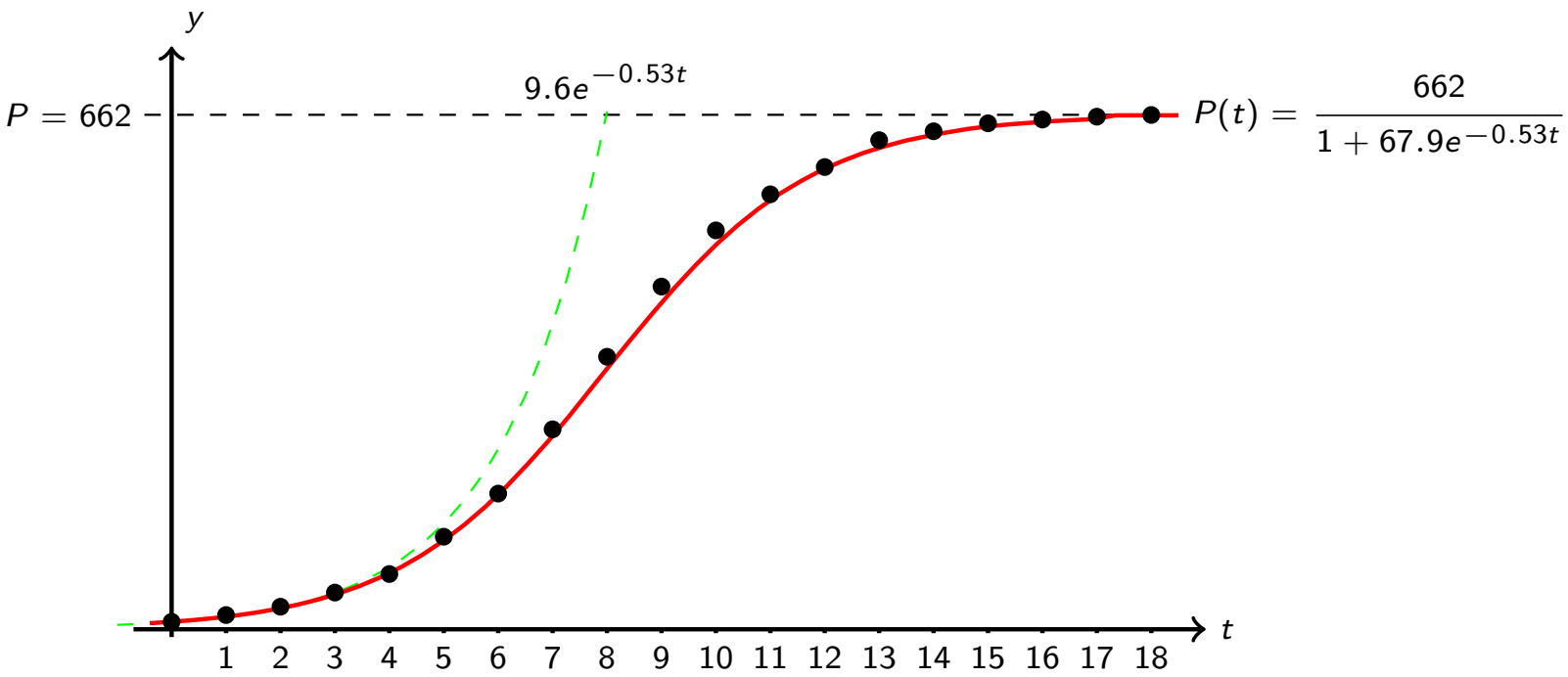
$$\frac{dP}{dt} = kP\left(1 - \frac{r}{k}P\right)$$

  has constant solutions $P = 0$ and $P = \frac{k}{r}$.
- The solution $P = \frac{k}{r}$ represents the theoretical maximum population
- Solution curves are sigmoidal (elongated "S" shaped) and approximately exponential when the population is small.



7.3 Applications of Differential Equations: Logistic Growth

- In spite of its simplicity, the logistic model has been "unreasonably successful".
- It describes the growth of many experimental populations remarkably well.



- Yeast culture example from *Raymond Pearl, Biology of Population Growth, 1925*.

7.3 Applications of Differential Equations: Newton's Law of Heating

- The rate at which an object cools (or heats) is proportional to the difference between its temperature T and the (constant) temperature T_s of its immediate surroundings. That is:

$$\frac{dT}{dt} = -k(T - T_s),$$

  where k is a positive constant.
- In applications, we can often determine $k > 0$ from experimental or observational data.
- Note $T = T_s$ is the only constant solution, $F(t) = -k$ and $G'(T) = 1$ are both well defined and continuous.

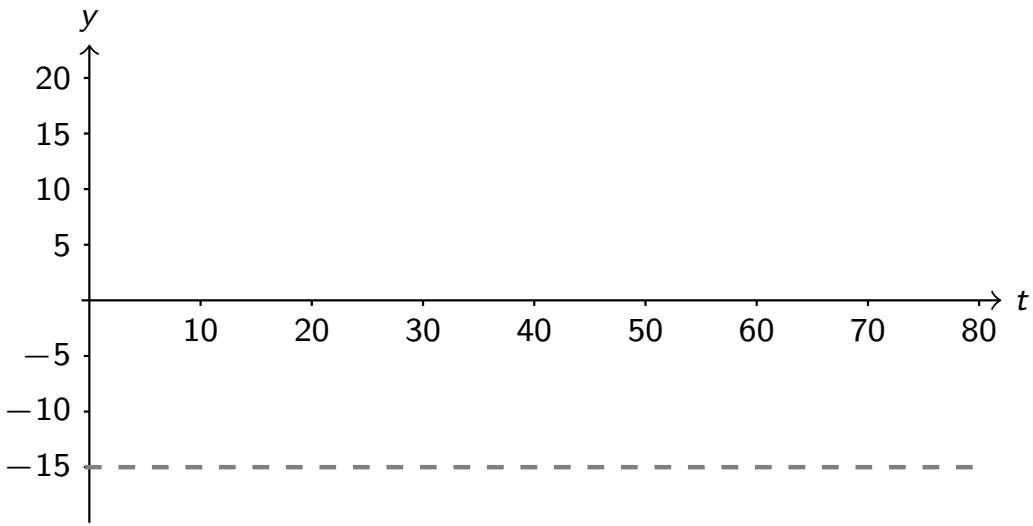
7.3 Applications of Differential Equations: Newton's Law of Heating

Example 7.25

A loaf of bread is placed in a freezer whose temperature is a constant $-15°$C. The bread obeys Newton's Law of Cooling. If the temperature T of the bread is initially $20°$C and it takes 20 minutes for it to drop to $10°$C, how long will it take for the bread to reach $0°$C?

7.3 Applications of Differential Equations: Newton's Law of Heating

7.3 Applications of Differential Equations: Newton's Law of Heating

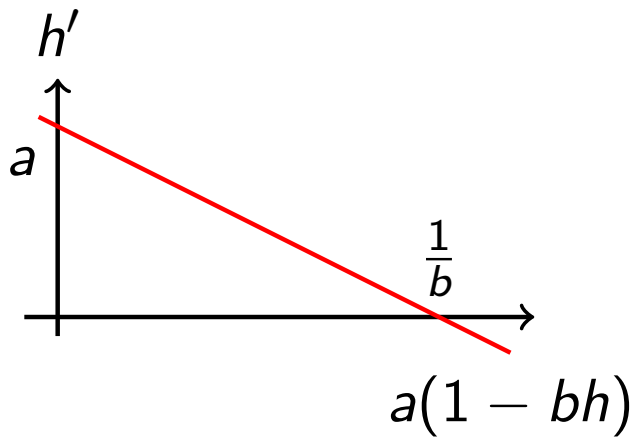

y
20
15
10
5
−5
−10
−15
10
20
30
40
50
60
70
80
t

7.3 Applications of Differential Equations: Tree Growth

- An initial value problem for growth of a tree

$$h'(t) = a(1 - bh(t)), \qquad h(0) = 0$$

  is derived by assuming that per unit time:
  1. Volume of photosynthetic nutrient production $\propto$ surface area.
  2. Number of cell deaths $\propto$ mass.
- a and b are positive constants, typically depending on the species.
- This model is relevant for h values in $[0, \frac{1}{b})$.

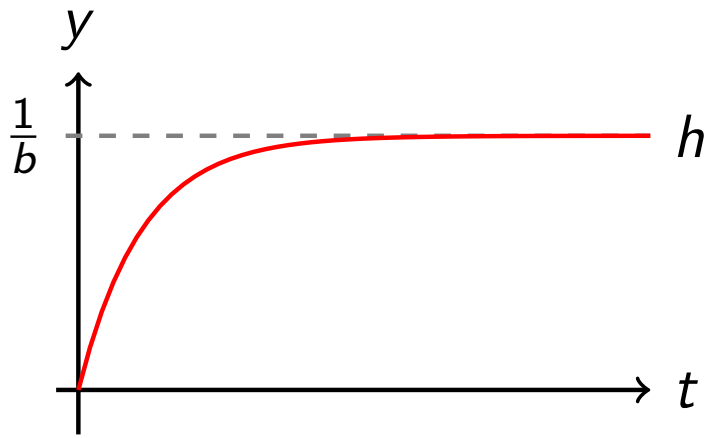


7.3 Applications of Differential Equations: Tree Growth

- Solving the initial value problem for tree growth gives

$$h(t) = \frac{1}{b}(1 - e^{-abt})$$

- As t goes to ∞ the height asymptotically approaches the growth limit $\frac{1}{b}$.



7.3 Applications of Differential Equations: Tree Growth

Example 7.26

Verify this solution and find the growth rate when $t = 0$.

7.3 Applications of Differential Equations: Tree Growth

Example 7.27

Suppose oak trees grow to an average maximum height of 20m and at 1 year of age their average height is 2m.

1. Find the average growth rate of an oak tree at birth.
2. Find the average height of an oak tree at age 2.